# Transfer Learning Report: ResNet50 on TF Flowers Dataset

## 1. Introduction

This project applies transfer learning using a pre-trained ResNet50 model on the TF Flowers dataset. The objective is to evaluate classification performance, apply fine-tuning strategies, and measure inference speed.

## 2. Dataset Description

The TF Flowers dataset consists of 5 classes of flower images.

Dataset split:

- 70% training
- 15% validation
- 15% testing

All images were resized to 224×224 and preprocessed using ResNet50-specific preprocessing.

## 3. Model Architecture (ResNet50)

ResNet50 is a deep convolutional neural network (50 layers) that uses residual connections to improve gradient flow and enable deeper networks to train effectively. It is pre-trained on ImageNet and adapted here using transfer learning by replacing the top classification layers.

## 4. Transfer Learning Setup

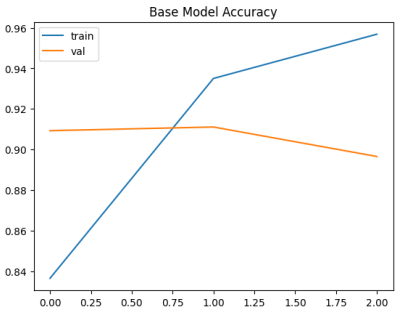
- Base model: ResNet50 (ImageNet weights)

- Base model initially frozen
- Classification head:
    - GlobalAveragePooling2D
    - Dense(128, ReLU)
    - Dense(5, Softmax)
- Loss function: Sparse Categorical Crossentropy
- Optimizer: Adam

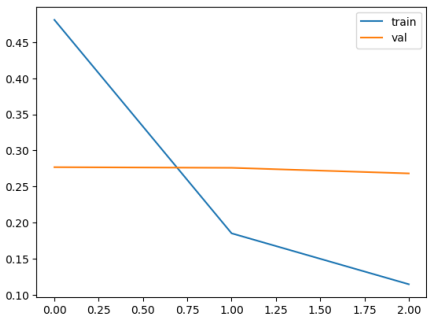
## 5. Base Model Training (Frozen)

Results:

- Training Accuracy: ~0.95 (final epoch)
- Validation Accuracy: ~0.89–0.91
- Test Accuracy: 0.905
- Base Accuracy Plot →




- Base Loss Plot →



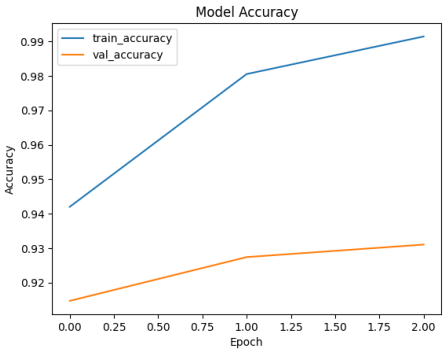

## 6. Fine-Tuning Strategy A (Last N Layers)

Setup:

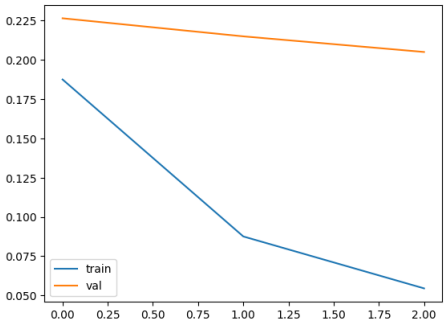
- Unfroze last 30 layers of ResNet50
- Learning rate: 1e-5

Results:

- Validation Accuracy: ~0.931
- Training Accuracy: ~0.99
- Strategy A Accuracy Plot →




- Strategy A Loss Plot →




## 7. Fine-Tuning Strategy B (Block-Based Unfreezing)

Setup:

- Unfroze conv5 block only (last residual block region)

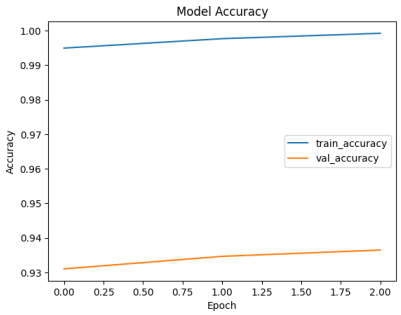
In Strategy B, only the last convolutional block (conv5) of ResNet50 was unfrozen. This is because early layers learn general features, while deeper layers learn task-specific features. Freezing early layers helps keep general knowledge and reduces overfitting while adapting the model to the dataset.

Results:

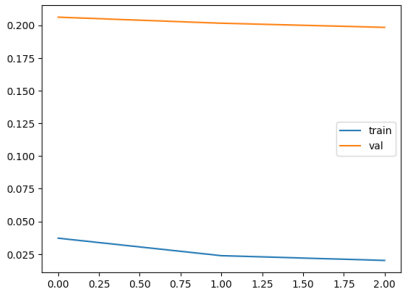
- Validation Accuracy: ~0.9365
- Training Accuracy: ~0.999

## Insert Plots Here:

- Strategy B Accuracy Plot →




- Strategy B Loss Plot →




# 8. Training Curves (Analysis)

- Rapid convergence within the first epoch
- Training accuracy approaches near-perfect values in fine-tuning stages
- Validation accuracy stabilizes around 0.93–0.94
- Loss decreases steadily and then plateaus

# 9. Inference Speed

Measured on a batch of test images:

- Batch inference time: 8.87 seconds (16 images)
- Approximate per-image time: ~0.55 seconds

This reflects the computational cost of ResNet50.

# 10. Unfreezing Strategies Analysis

## Strategy A

- Simple implementation
- Faster adaptation
- Slightly more prone to overfitting

## Strategy B

- Architecture-aware (block-based)
- Slightly better validation accuracy
- More stable training behavior

# 11. Limitations

- Small dataset size increases overfitting risk
- Only ResNet50 evaluated (MobileNet not included in this report)
- Inference time depends on hardware (CPU/GPU differences)

# 12. Conclusion

ResNet50 achieved strong performance on the TF Flowers dataset with a final test accuracy of approximately 0.905 and validation accuracy up to 0.936 after fine-tuning.

Fine-tuning significantly improved performance compared to the frozen base model. Strategy B (block-based unfreezing) provided slightly better results than Strategy A, though both approaches show mild overfitting due to dataset limitations.

Overall, ResNet50 provides high accuracy at the cost of higher computational complexity and slower inference speed.